

Quiz1 2025
Biophysics(BIOP)

Total Marks:20

Time: 45 minutes

Section-A (MCQ 1*10 Marks)

1. A Nucleoside differs from a nucleotide in the following way:
 - a. Nucleotide has a phosphate group; nucleoside does not
 - b. Nucleotide lacks a sugar; nucleoside has a sugar
 - c. Nucleotide lacks a nitrogenous base; nucleoside has a base
 - d. Nucleotide lacks hydrogen bonding; nucleoside has hydrogen bonding
2. The basic principle of X-Ray Crystallography is based on:
 - a. Diffraction of X-rays by protons in a crystal lattice.
 - b. Absorption of X-rays by crystal lattice.
 - c. Diffraction of X-rays by electrons in a crystal lattice.
 - d. Emission of X-rays from atoms
3. The 2017 Nobel Prize in Chemistry was awarded for the development of which technique?
 - a. X-ray crystallography
 - b. Cryo-electron microscopy (Cryo-EM)
 - c. Nuclear magnetic resonance (NMR) spectroscopy
 - d. Mass spectrometry
4. The parameter ' σ ' in the Lennard-Jones potential best corresponds to which of the following definitions?
 - a. The depth of the energy well.
 - b. The distance where the potential energy $V(r)$ is zero.
 - c. The distance of maximum repulsion.
 - d. The sum of the minimum van der Waals radii from the table.
5. Identify the correct bond representation of torsion angle ϕ (phi) and ψ (psi) in the proteins?
 - a. $\phi(\text{C-N-CA-C})$ and $\psi(\text{N-CA-C-N})$
 - b. $\phi(\text{C-N-CA-N})$ and $\psi(\text{C-N-CA-N})$
 - c. $\phi(\text{N-CA-N-C})$ and $\psi(\text{CA-N-CA-C})$
 - d. $\phi(\text{CA-C-N-CA})$ and $\psi(\text{N-C-N-CA})$

6. How does the hydrogen bonding in an alpha-helix differ from that in a beta-sheet?
- a. Alpha-helices uses side-chain donors/acceptors, while beta-sheets use the protein backbone.
 - b. ~~Alpha-helices form intra-chain H-bonds, while beta-sheets form inter-chain H-bonds.~~
 - c. Beta-sheet form inter-chain H-bonds, while alpha-helices form intra-chain H-bonds.
 - d. Alpha helices and beta sheets both form inter-chain H-bonds
7. During protein synthesis, the polypeptide chain is elongated in which direction?
- a. From the C-terminus to the N-terminus, adding new amino acids to the carboxyl group.
 - b. From the N-terminus to the C-terminus, adding new amino acids to the carboxyl group.
 - c. From the C-terminus to the N-terminus, adding new amino acids to the amino group.
 - d. From the N-terminus to the C-terminus, adding new amino acids to the amino group.
8. Which two discoveries were most critical for Watson and Crick's determination of the double-helix structure of DNA?
- a. Pauli exclusion principle and London dispersion forces
 - b. Anfinsen's dogma and Levinthal's paradox
 - c. Chargaff's rules and X-ray diffraction patterns
 - d. Alpha-helix H-bonding and beta-sheet formation
9. A protein is identified as being insoluble, highly regular in structure, and forms strong aggregates. It is most likely classified as a:
- a. Globular protein
 - b. Fibrous protein
 - c. Membrane protein
 - d. Enzymatic protein
10. The Lennard-Jones potential describes interactions between atoms. Which of the following correctly identifies its two components and the forces they represent?
- a. Attractive term ($-1/r^6$): van der Waals forces; Repulsive term ($+1/r^{12}$): Pauli repulsion
 - b. Attractive term ($-1/r^6$): Ionic forces; Repulsive term ($+1/r^{12}$): Covalent bonding
 - c. Attractive term ($-1/r^{12}$): van der Waals forces; Repulsive term ($+1/r^6$): Pauli repulsion
 - d. Attractive term ($-1/r^6$): Hydrogen bonding; Repulsive term ($+1/r^{12}$): van der Waals forces

Section-B (2*5=10 mark)

1. The Buckingham and Lennard-Jones potentials both model van der Waals interactions. Justify why the more accurate Buckingham potential is often replaced by the Lennard-Jones potential in molecular simulations.

Ans: The Buckingham potential uses an exponential term ($A\exp(-Br)$) to model quantum repulsion. Calculating exponential functions is computationally expensive and slow for a computer, especially when done for millions of atom pairs in a simulation. [1 mark]

Computational Efficiency: The Lennard-Jones potential uses a $1/r^{12}$ term for repulsion. This is extremely efficient to compute because the attractive $1/r^6$ term must already be calculated for the dispersion force. The repulsive term can be found with just one additional multiplication: $(1/r^6) * (1/r^6)$. [1 mark]

2. Justify why α -helices are considered more flexible yet more stable compared to β -sheets.

Answer: α -helix is generally more stable, it has regular intra-chain H-bonds ($i \rightarrow i+4$) that satisfy backbone donors/acceptors within the same strand, producing a compact, low-energy conformation. β -sheets depend on inter-strand H-bonds and proper strand alignment; they can be very stable when many strands form an extended network (e.g., amyloid), but individual strands are more susceptible to edge exposure and solvent, so β -sheets are not intrinsically more stable than helices in isolation.

3. Describe the utility of a Ramachandran plot in the process of validating a protein's three-dimensional model. What specific structural features does it help to identify?

Ans: A Ramachandran plot displays the allowed dihedral angles (ϕ and ψ) of amino acid residues in a protein. It is used to validate protein 3D models by checking whether residues fall within favored regions. It helps identify specific secondary structures like α -helices and β -sheets, and highlights residues in disallowed regions that may indicate errors in the protein model.

4. A short peptide has 3 residues. Its Φ and Ψ angles are sampled in 90-degree increments.

- a. Calculate the number of possible conformations. (1.5 mark)

Ans. $360^\circ/90^\circ = 4$ states per angle. [0.5 mark]

$4^2 = 16$ conformations per residue. [0.5 mark]

For 3 residues: $16^3 = 4096$ possible conformations. [0.5 mark]

- b. Name the paradox that this type of calculation illustrates? (0.5 mark)

Ans. **Levinthal paradox**: random search of all possible conformations to find the native one

5. Which conformation is generally not observed in peptide bonds and why? Give two reasons. State the exception.

Ans. **Cis** conformation is extremely rare. [0.5 mark]

Reason:

Steric clash: In the cis geometry, the two adjacent C α substituents (and their side chains) lie on the same side and collide sterically, which raises the energy compared with the trans arrangement. [0.5 mark]

Peptide resonance & planarity: The peptide bond has strong partial double-bond character (resonance), fixing it planar and making the lower-energy, less-crowded trans orientation strongly preferred. [0.5 mark]

Exception: proline where the cyclic side chain reduces steric hindrance, allowing the cis conformation to occur more frequently [0.5 mark]